COMMUNICATION INTERFACE

Smart Meter

Data Concentrator

LoRaWAN Communication

- Establish a gateway within the operational area for effective LoRaWAN communication with a range of up to 5km.
- Instead of assigning individual unique IDs to each smart meter, use their DevEUIs to identify and associate data with specific devices.
- Store these DevEUIs in a map data structure, using the residence owner's names as keys.
- This approach simplifies pricing calculations for each home by easily associating data with the respective residences.

RS485 Cables

- Inverters and smart meters connect via RS485 cables.
- Benefits of communication between inverter and smart meter in solar power systems:
  1. Enhances Self-Consumption: Inverters can align their operation with building electricity usage, optimizing self-consumption.
  2. Grid Interaction: Facilitates accurate measurement and reporting of surplus solar energy sent to the grid, crucial for billing in regions with net metering or feed-in tariffs.

Wifi Network

- All houses are equipped with modems for internet connectivity.
- Inverters are connected to the house's Wi-Fi using a username and password.
- A software application is used to display the parameters of the inverters and receive error signals.
- Data concentrators process information from up to 1024 smart meters locally.
- Processed data is then uploaded via optic fiber for rapid connection to either Google Cloud or AWS.
- Large datasets are stored and analyzed using AI models to optimize pricing units.
- The data concentrator also monitors inverter performance over the network.
- In case of errors, the data concentrator triggers a remote shutdown to prevent grid-wide issues.

Inverter

Software Application or Website

User

- A software application or or website is used to display the parameters of the inverters and receive error signals.
- In case of errors, the data concentrator triggers a remote shutdown to prevent grid-wide issues rather than the user since the user won't be continously checking the application.